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A Comparative Review of Pore-Scale Nitrogen, Carbon Dioxide, Hydrogen, and Methane Transport, Hysteresis, and Ostwald Ripening in Storage Reservoirs

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Abstract

This review systematically compares the multiphase flow and transport behavior of nitrogen (N_2), carbon dioxide (CO_2), hydrogen (H_2), and methane (CH_4) in porous rocks, including sandstones and carbonates. It focuses on examining variations in hysteresis, Ostwald ripening, wettability, mass transport, and trapping of each gas at the pore-scale. The objective is to understand how different these gases interact with and move through reservoir rocks, offering a comparative perspective for underground gas storage applications. Multiple experimental approaches have been reviewed to assess the behavior of different gases in porous rocks. Pore-scale imaging and core-flooding experiments were evaluated to investigate gas distribution and transport within rock samples. Techniques such as X-ray micro-computed tomography have facilitated the visualization of fluid distribution and *in situ* monitoring of saturation changes. Core flooding experiments, conducted under representative reservoir conditions, provided measurements of gas breakthrough, relative permeability, and residual trapping. These methodologies enable a comparison of N_2, CO_2, H_2, and CH_4, offering valuable insights into their transport and trapping behaviors in reservoir rocks. Pore-scale transport phenomena reveal distinct behaviors among the different gases due to differences in molecular properties. H_2, as the smallest and fastest-diffusing gas, shows enhanced redistribution via Ostwald ripening. Over time, this leads to significant cluster coarsening and reduced trapping hysteresis. In contrast, CH_4 and N_2 exhibit minimal redistribution and maintain stronger hysteresis due to slower diffusion. CO_2 also shows pronounced Ostwald ripening, with trapped clusters rapidly coarsening within hours. Hysteresis behavior varies by gas. N_2 shows slightly weaker, yet still significant, relative permeability hysteresis. CO_2, especially in carbonates, can alter wettability which can reduce the amount of capillary trapping and hence hysteresis in flow properties. H_2 may also exhibit decreased hysteresis over cycles, but due to a different mechanism, as redistribution due to Ostwald ripening reconnects previously trapped gas. Adsorption data indicate CO_2 strongly adsorbs to mineral surfaces, forming dense layers, whereas H_2 shows negligible adsorption. CH_4 and N_2 have intermediate tendencies. Diffusion results shows that H_2 diffuses fastest while CO_2 diffuses slowest. Flooding tests show that all gases experience significant residual trapping, with

slightly lower residual H_2 due to its high mobility. These findings highlight the critical roles of wettability, pore structure, and gas properties in determining subsurface gas trapping and transport dynamics. This study highlights H_2's unique behavior in subsurface environments due to its high diffusion rate and low adsorption, enabling Ostwald ripening and potential hysteresis change, unlike CH_4 and N_2. CO_2's dual role in modifying wettability and undergoing rapid Ostwald coarsening is also emphasized. These findings contribute new insights into gas-specific transport and trapping dynamics, offering valuable insight for optimizing underground storage strategies.

Introduction

To mitigate the continued use of fossil fuels while addressing climate change, one viable approach is the implementation of carbon dioxide capture and storage (CCS). This technology has been recognized as a feasible means of limiting global temperature rise (Alcalde et al., 2018; Bellamy & Geden, 2019). An alternative strategy involves replacing fossil fuels with clean, renewable energy sources. However, the intermittent and unpredictable nature of renewable energy generation poses a significant limitation to its reliability as a primary energy source. To address this intermittency, excess energy from renewables can be converted into H_2 via electrolysis and stored underground, a process known as underground hydrogen storage (UHS). Stored hydrogen can then be withdrawn when the energy demand peaks. Particularly, the large capacity of subsurface storage and its long-term usage makes UHS a suitable solution for balancing fluctuations in renewable energy supply (Shaqsi et al., 2020; Heinemann et al., 2021). In addition to its energy balancing potential, UHS is considered safer than surface H_2 storage, as it avoids direct contact between H_2 and atmospheric oxygen—thereby reducing the risk of flammability (Tarkowski, 2019; Zivar et al., 2021).

Gases such as nitrogen (N_2), carbon dioxide (CO_2), hydrogen (H_2), and methane (CH_4) can behave differently in porous media due to differences in their density, solubility, and diffusivity. Understanding these distinctions can help develop effective underground gas storage strategies. For instance, CO_2 is injected for carbon sequestration and trapping, CH_4 occurs naturally in subsurface reservoirs, N_2 can serve as a cushion gas, and H_2 is gaining attention as a clean energy source stored underground. A key subsurface transport process relevant to UHS is Ostwald ripening, which is the diffusion of gas through the brine phase from regions of high capillary pressure to those of lower pressure within the pore network (Hematpur et al., 2023; Zhang et al., 2023). This phenomenon leads to the redistribution of hydrogen, with gas accumulating in larger pores while diminishing in smaller ones (de Chalendar et al., 2018; Hematpur et al., 2023; Zhang et al., 2023). This has been demonstrated at laboratory-scale experiments where significant gas rearrangement was observed over timescales of hours to days (Dokhon et al., 2025, Goodarzi et al., 2025). When considered at the field scale, such gas redistribution due to Ostwald ripening can occur over a timescale of several years (Blunt, 2022).

X-ray imaging has been widely adopted for observing pore structures and the distribution of fluids within pore systems in three dimensions at micrometer-scale resolution (Wildenschild & Sheppard, 2013). Among the most advanced techniques currently available is micro-computed tomography (μ-CT), a dependable lab-based tool for imaging rock samples. In recent years, μ-CT has been extensively applied to analyze various properties at the pore scale, including porosity profile, saturation distribution, relative permeability, local capillary pressure, and *in situ* contact angle (Alhammadi et al., 2017; Herring et al., 2017; Scanziani et al., 2018). Numerous studies have explored H_2 injection into porous media (Yekta et al., 2018; Heinemann et al., 2021; Iglauer et al., 2021; Jha et al., 2021; Higgs et al., 2022; Jangda et al., 2023; Thaysen et al., 2023; Goodarzi et al., 2024) to evaluate the initial gas saturation. In water-wet systems, for instance, a larger volume of gas tends to become trapped by water due to the capillary forces, a feature advantageous for CO_2 storage but unfavorable for H_2 storage (Yekta et al., 2018; Hashemi et al., 2021b).

Several studies have examined the Ostwald ripening phenomenon in the context of CO_2, air, and natural gas systems (de Chalendar et al., 2018; Singh et al., 2022; Gao et al., 2023). This process is particularly relevant for H_2 storage due to H_2's high diffusion coefficient in water (Blunt, 2022). In one study H_2 and N_2 were injected into Bentheimer sandstone and visualized using micro-computed tomography: over time, notable rearrangements in the trapped H_2 clusters were observed, with smaller bubbles shrinking and larger ones growing—behavior more pronounced compared to N_2 (Zhang et al., 2023). Furthermore, H_2 can rearrange to form a large connected ganglion over a length of 3 cm (Goodarzi et al., 2024; Dokhon et al., 2025). X-ray imaging techniques have also been employed to quantify H_2 saturation in brine-filled sandstone formations (Jha et al., 2021; Higgs et al., 2022). These studies reported an initial gas saturation of up to 65% after primary drainage, and around 41% residual gas saturation after imbibition. Moreover, medical X-ray CT scanning of Berea sandstone demonstrated that H_2 trapping at the centimeter scale is influenced by viscous, capillary, and gravitational forces (Thaysen et al., 2023).

Higgs et al. (2023) conducted a comparative study on H_2, N_2, and CH_4, focusing on relative permeability, interfacial tension, saturation distribution, and wettability. While interfacial tension and wettability showed similar results across the three gases, N_2-brine systems exhibited higher relative permeability than those involving H_2 and CH_4. Capillary pressure measurements were also comparable for all gases tested; however, the study recommended further investigation under high-pressure and high-temperature conditions. Additionally, AlZaabi et al. (2025) compared N_2, CO_2, and H_2 with respect to wettability, pore occupancy, capillary pressure, and gas ganglia volume distribution. Their findings showed that H_2 had significant gas rearrangement due to Ostwald ripening during storage, while N_2 showed the least rearrangement and CO_2 exhibited intermediate behavior. In terms of ganglia volumes, H_2 formed the largest gas ganglion, while N_2 had no substantial gas rearrangement during storage, and CO_2 formed connected ganglia of moderate volume.

Understanding displacement and mass transfer processes at the pore scale remains limited for UHS. Improved insights at this scale can enhance the accuracy of reservoir-scale simulations, which are essential for assessing the feasibility of UHS projects. This paper reviews the available experimental studies to present a comparative review of N_2, CO_2, CH_4, and H_2, emphasizing the distinct behavior of H_2 in subsurface environments. Due to its high diffusivity and low adsorption, H_2 exhibits pronounced Ostwald ripening and potential hysteresis effects, which are less evident in CH_4 and N_2. The study also highlights CO_2's unique dual influence—altering wettability and promoting rapid Ostwald ripening. Overall, these insights enhance our understanding of gas-specific transport and trapping mechanisms, providing a foundation for more effective underground storage strategies.

Experimental Methods for Determination of Pore-Scale Behavior

A range of experimental methods has been used to study how different gases behave in porous rocks. Pore-scale imaging and core-flooding tests have been used to analyze gas movement and distribution. Tools including μ-CT have enabled visualization of fluid arrangement and real-time tracking of saturation changes. Experiments, carried out under reservoir conditions, have yielded data on gas breakthrough, relative permeability, and residual trapping. These techniques allow for direct comparison of N_2, CO_2, H_2, and CH_4, providing important insights into their movement and trapping characteristics in reservoir formations. This review focuses on experiments where different gases were directly compared in the same rock. A summary of the experimental studies is provided in Table 1.

Table 1—Studies that directly compare different gases in the same rock system.

Reference	Gases	Rock	Methodology	Summary
Higgs et al. (2023)	H_2, CH_4, N_2	Bentheimer sandstone	µ-CT	Investigated and compared petrophysical and two-phase flow properties of H_2-brine, CH_4-brine, and N_2-brine systems under *in situ* conditions.
Zhang et al. (2023)	H_2, N_2	Bentheimer sandstone	µ-CT	Compared Ostwald ripening of H_2 and N_2 in sandstone rock during injection and withdrawal cycles.
AlZaabi et al. (2025)	N_2, CO_2, H_2	Carbonate	µ-CT	Directly compared the pore-scale behavior of N_2, CO_2, and H_2 in a carbonate reservoir rock, focusing on wettability, pore occupancy, connectivity, and Ostwald ripening.
Al-Yaseri et al. (2022)	H_2, N_2	Fontainebleau sandstone	Core flooding	Quantified initial and residual gas saturations of H_2 and N_2 in sandstone rock under shallow reservoir conditions.
Lysyy et al. (2022)	H_2, N_2	Berea sandstone	Core flooding	Measured hysteresis in H_2-brine relative permeability in sandstone under shallow reservoir conditions.
Rezaei et al. (2022)	H_2, N_2, CH_4	Sandstone and Carbonate	Core flooding	Investigated H_2 flow in sandstone and carbonate rocks when coexisting with brines of different salinity under reservoir conditions using unsteady-state drainage experiments and compared the results to N_2 and CH_4.
Iglauer et al. (2021)	H_2, CO_2	Sandstone	Tilted plate contact angle	Investigated H_2 and CO_2 wettability and interactions with sandstone under reservoir conditions.
Mirchi et al. (2022)	H_2, CH_4	Berea sandstone (oil-wet)	Sessile drop contact angle method	Measured the interfacial tensions of brine in contact with H_2–CH_4 mixtures and determined contact angles formed between these gas mixtures, brine, and oil-wet rock.
Song et al. (2025)	H_2, N_2, CH_4	Salt rock and mudstone	Manometric, volumetric and gravimetric	Investigated H_2, N_2, and CH_4 diffusion through rock and how this diffusion contributes to gas leakage in underground gas storage systems.
Alanazi et al. (2023)	H_2, CO_2, CH_4, N_2	Organic-rich carbonate	X-ray diffraction, scanning electron microscopy, thermogravimetric analysis, and spectroscopic techniques	Investigated the adsorption characteristics of H_2, CO_2, and CH_4 on organically rich carbonate rock to evaluate underground storage suitability.
Al-Yaseri et al. (2023)	H_2, CH_4	Organic-rich carbonate	Exposure experiment, interfacial tension measurement and pendent drop method	Investigated the interfacial tension and contact angle of H_2 and CH_4 with brine and organic-rich carbonate rock.

Ostwald Ripening and Gas Redistribution

Gases within small pores have high local capillary pressure which, by Henry's law, create a high dissolved gas concentration at the interface. Variations in the local capillary pressure therefore create a gradient of dissolved gas concentration that drives the rearrangement process (Hematpur et al., 2023). In free fluids, this process leads to gas consolidating into a single large bubble at equilibrium. In porous media, however, multiple equilibrium sites can exist (i.e., local equilibria), causing smaller gas bubbles to shrink as gas diffuses toward larger gas clusters (de Chalendar et al., 2018; Hematpur et al., 2023). This mechanism can enable the movement of previously trapped gas and help reduce hysteresis effects (Adebimpe et al., 2024). Despite H_2's very low solubility in water, its small molecular size and high diffusivity enable rapid equilibration of dissolved H_2 within the brine. Pore-scale imaging experiments clearly illustrate this Ostwald ripening behavior. For example, μ-CT observations of water-saturated sandstone cores revealed that following H_2 injection and subsequent water imbibition, numerous small trapped H_2 bubbles vanished while larger H_2 clusters expanded—occurring without any net loss in total gas volume (Zhang et al., 2023; Goodarzi et al., 2024; AlZaabi et al., 2025; Dokhon et al., 2025). When compared to N_2 and CO_2, H_2 shows the most gas redistribution followed by CO_2 then N_2 (AlZaabi et al., 2025). Since Ostwald ripening is a diffusion driven process, it can be assumed that CH_4 can demonstrate an Ostwald ripening between N_2 and H_2 based on the gases diffusion coefficient (Song et al., 2025). Figure 1 demonstrates Ostwald ripening for different gases at the pore scale.

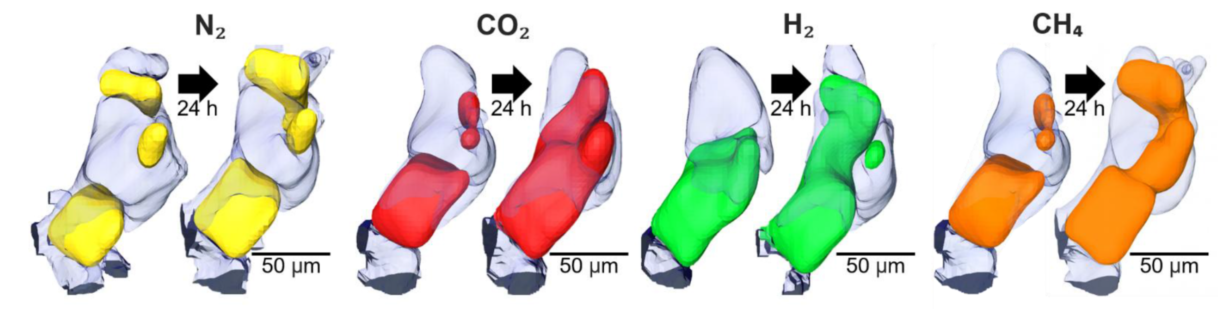

Figure 1—The effect of Ostwald ripening on N_2, CO_2, H_2 and CH_4 after 24 hours. The images for N_2, CO_2 and H_2 were determined experimentally from a previous study (AlZaabi et al., 2025). The speed and extent of Ostwald ripening from highest to lowest is in the order H_2>CH_4>CO_2>N_2.

In the context of UHS, Ostwald ripening can enable multiple H_2 injection and withdrawal cycles with minimal gas trapping. At the millimeter scale, this process redistributes stored H_2 over timescales of hours to days. However, at the field scale, H_2 redistribution occurs over much longer periods—ranging from years to millions of years (Blunt, 2022). Consequently, for storage cycles lasting only a few months, local equilibrium occurs at the centimeter scale, which is insufficient to significantly alter the overall H_2 distribution. While the influence of Ostwald ripening on H_2 rearrangement is limited to just a few centimeters, this will impact macroscopic flow properties, such as relative permeability and capillary pressure such that H_2 flow at the micrometer scale leads to reduced trapping and less hysteresis—unlike the multiphase flow behavior observed in oil and gas production. This suggests that, during H_2 storage, a smaller volume of cushion gas may be sufficient for effective injection and withdrawal operations (Hematpur et al., 2023).

Relative Permeability Hysteresis

Hysteresis in pore-scale models has been investigated: a critical factor in UHS is the proportion of H_2 that can be recovered during withdrawal, as some H_2 becomes trapped in the reservoir brine (Hashemi et al., 2021a). The extent of H_2 trapping in porous media is primarily determined by rock wettability and pore structure. For instance, in a strongly water-wet system such as an aquifer, a significant residual gas saturation is expected. In conventional oilfield operations, hysteresis is typically assumed to result from the displacement of two completely immiscible fluids, with hydrocarbon dissolution considered negligible. However, this assumption does not hold for H_2 or CO_2 (Hematpur et al., 2023). The degree and nature of hysteresis varies significantly between gases, influencing both the physical trapping mechanisms and the long-term feasibility of underground gas storage strategies. For CO_2, hysteresis effects are present but less pronounced than for H_2 (Zhong et al., 2024). It has been found that brine relative permeability hysteresis in an H_2-brine system is much stronger than in a CO_2-brine system, largely due to differences in contact angle hysteresis. This disparity implies that in UHS, intermittent injection strategies must be carefully managed, as strong hysteresis could significantly reduce recovery efficiency. In the case of H_2, multiple studies highlight the substantial impact of hysteresis on storage performance. Bahrami et al. (2023) reported that hysteresis in relative permeability can reduce H_2 recovery factors by 16–25%, highlighting the importance of accurate hysteresis modeling in reservoir simulations. Experimental studies further confirm this. For example, Lysyy et al. (2022) observed strong hysteresis in H_2 and water relative permeabilities in Berea sandstone, with the gas relative permeability higher during drainage and the water relative permeability higher during imbibition. This behavior was linked to residual gas trapping and contact angle hysteresis, both of which reduce the mobility of H_2 in subsequent production cycles.

Other mechanisms, such as Ostwald ripening, tends to reduce the hysteresis effect (Zhang et al., 2023; Goodarzi et al., 2024). Zhang et al. (2023) showed that in Bentheimer sandstone, H_2 saturation was reduced due to Ostwald ripening. In contrast, when N_2 was stored under similar conditions in Bentheimer sandstone, no significant reduction in saturation was observed, and the configuration remained stable over 12 hours, indicating greater stability and lower susceptibility to Otwald ripening (no reduction in hysteresis). Additionally, Dokhon et al. (2025) demonstrated that hydrogen rearrangement during storage and after spontaneous imbibition significantly improved gas connectivity compared to rearrangement after brine injection. Overall, these findings suggest that the extent and causes of hysteresis differ sharply between gases due to variations in interfacial properties, wettability, and processes such as Ostwald ripening. For UHS, Ostwald ripening reduces hysteresis and residual saturation, which in favorable for repeated cycles of injection and withdrawal. In contrast, CO_2 and N_2 exhibit comparatively more stable hysteresis behavior, with potentially more trapping which is favorable for long-term sequestration. However, this ignores the effect of wettability which is discussed next.

Wettability and Adsorption

Wettability is a key factor controlling multiphase flow and trapping in geological formations, and thus directly influences the efficiency of underground gas storage. Across multiple studies (Table 2), distinct wettability trends emerge, driven by different factors such as mineralogy, brine salinity, temperature, pressure, and rock-fluid interactions over time.

Most experiments have been performed under water-wet conditions. In carbonate rocks with high-salinity brines (44.3% total salinity), AlZaabi et al. (2025) reported strongly water-wet conditions (contact angles <20°) for N_2, CO_2, and H_2. Similar water-wet behavior was observed by Higgs et al. (2023) for Bentheimer sandstone with deionized water, with contact angles for H_2 (44°), CH_4 (46°), and N_2 (45°) all indicating strong water affinity of the rock surface. Iglauer et al. (2021) also found H_2 to be water-wet (contact angle <76.7°) in sandstones, though they noted that CO_2 exhibited slightly higher contact angles profile with pressure up to 25 MPa (<95°) than H_2, a difference likely linked to CO_2's higher interfacial reactivity and solubility in brines. This is supported by Eyitayo et al. (2024) where CO_2 tends to alter wettability in sandstones and carbonates towards more CO_2-wet conditions (higher contact angles). Comparative wettability differences between gases are consistently reported. Zhong et al. (2024) observed larger contact angles for CO_2 than for H_2, suggesting a relatively less water-wet state for CO_2. Time-dependent wettability shifts are seen where Zhang et al. (2023) found that in Bentheimer sandstone with KI brine, initial contact angles indicated weakly water-wet conditions (60-70°). After 12 hours, N_2's contact angle remained stable, whereas H_2's increased by approximately 10°, suggesting that H_2-brine-rock interactions over time can promote a shift toward more intermediate wettability, potentially impacting long-term injectivity and recovery.

Table 2—Summary of experimental wettability measurements for different gases.

Study	Gas	Rock	Temperature (K)	Pressure (MPa)	Brine total salinity (wt%)	Wettability (contact angle, °)
Al-Yaseri et al. (2023)	H_2	Organic-rich carbonate (TOC ≈14 wt%.)	348	3.45–10.34	10 (NaCl)	~94–106
	CH_4					~131–149
AlZaabi et al. (2025)	CH_4	Carbonate	333	8	44.3	~131–149
	CO_2					Water-wet (~20)
	H_2					Water-wet (~15)
Higgs et al. (2023)	H_2	Bentheimer sandstone		2.13	0 (DI water)	Water-wet (44)
	CH_4					Water-wet (46)
	N_2					Water-wet (45)
Iglauer et al. (2021)	H_2	Sandstone	296-343	0.1-25	10 (NaCl)	Water-wet θ_a (33-76.7) and θ_r (28-70.7)
	CO_2					Water-wet θ_a (65-95) and θ_r (63-85)
Mirchi et al. (2022)	H_2	Berea sandstone (oil-wet)	295 313 333	6.9	2	62.37 ± 2.07 63.72 ± 1.94 61.16 ± 2.48
	CH_4					70.73 ± 1.62 71.86 ± 3.04 69.89 ± 3.47
	H_2	Indiana limestone (oil-wet)				65.18 ± 4.2 64.69 ± 3.38 63.10 ± 3.38
	CH_4					77.57 ± 0.42 76.76 ± 0.49 74.85 ± 0.77
Zhang et al. (2023)	H_2	Bentheimer sandstone	298	1	3.4 (KI)	Weakly water-wet (60–70). After 12 h, contact angle remained stable for N_2 but increases by ~10 for H_2.
	N_2					
Zhong et al. (2024)	CO_2					Contact angle larger for CO_2 than H_2.
	H_2					

Oil-wet systems reveal significant wettability contrast between gases. In Berea sandstone, Mirchi et al. (2022) observed lower contact angles for H_2 (61-64°) than CH_4 (70-72°), implying that H_2 may exhibit stronger adhesion to the brine phase than methane in oil-wet media. The same trend was observed in Indiana limestone, with H_2 (63–65°) being more water-wet than CH_4 (75-78°). In organic-rich carbonates, Al-Yaseri et al. (2023) measured contact angles exceeding 90° for both H_2 (~94-106°) and CH_4 (~131-149°), indicating intermediate to gas-wet behavior. Here, CH_4 was significantly more gas-wet than H_2, reflecting the influence of total organic content (TOC) on interfacial interactions, where hydrophobic organic matter preferentially stabilizes nonpolar gases. H_2 generally displays strong to moderate water-wetness in most mineralogical contexts but can shift toward intermediate wettability under certain brine compositions or prolonged exposure. In contrast, CO_2's wettability is more variable, while N_2 tends to remain stable and strongly water-wet. Overall, in water-wet conditions, all the gases display similar wettability, but the wettability varies significantly with different gases in systems that are more oil-wet.

Figure 2 demonstrate contact angle trends with temperature where green, black, blue and red corresponds to H_2, CH_4, N_2 and CO_2 respectively. H_2 on water-wet sandstone shows a clear increase in contact angle from ~25° at 296 K to ~45° at 343 K, whereas oil-wet systems (e.g., Berea sandstone and Indiana limestone) exhibit only weak, slightly decreasing, temperature dependences. CH_4 generally yields the largest angles, remaining ~70–80° in oil-wet rocks and reaching >100° in organic-rich carbonates, while CO_2 and N_2 contact angles are intermediate and maybe rock system-dependent (e.g., CO_2 low on water-wet carbonate but moderate to high on sandstone). Wettability trends with pressure are compared in Figure 3 (same color coordination for each gas as Figure 2) where contact angle increases monotonically with pressure for H_2,

CO_2 and CH_4 (typical rises of ~10–20° between ~5 and 25 MPa for H_2 and CH_4) while CH_4 is consistently high overall then the other gases. This implies that the temperature and pressure mostly increase contact angle with the exception of CO_2 contact angle decreasing with higher temperature. H_2 stays relatively water-wet in sandstones which is favorable for storage. CH_4 can reach higher contact angle which can restrict the gas mobility—especially in oil-wet (organic-rich) carbonates.

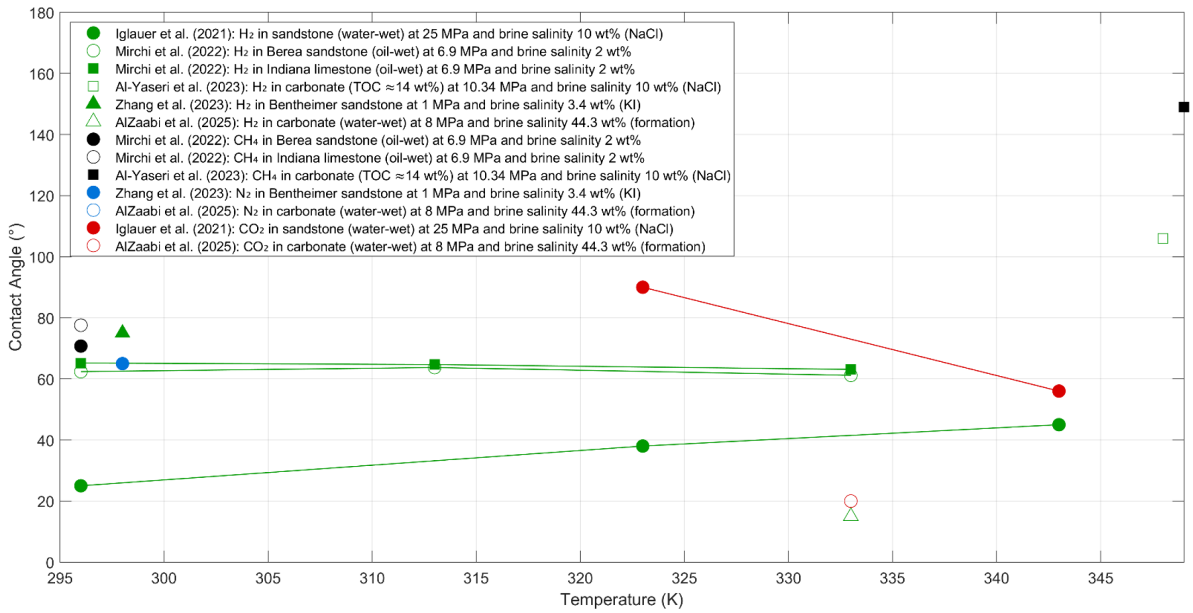

Figure 2—Contact angle as a function of temperature for different gases in different brine/rock systems.

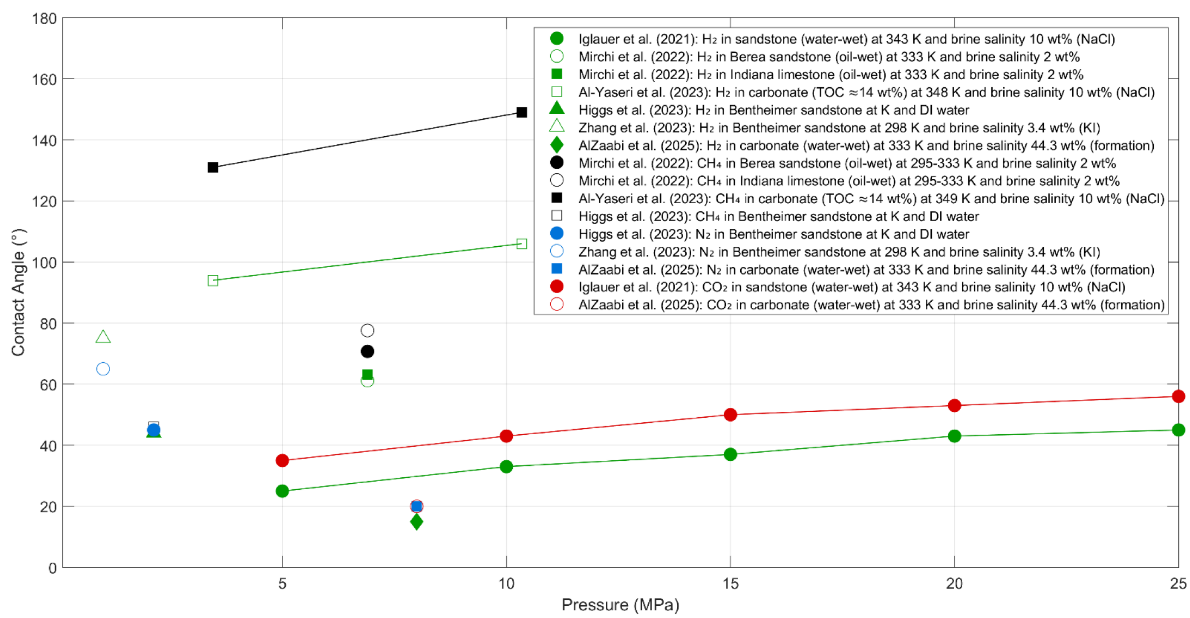

Figure 3—Contact angle as a function of pressure for different gases in different brine/rock systems.

Gas Transport and Trapping

Gas transport and trapping behavior in porous media are governed by a complex interplay of fluid properties (viscosity, density, solubility), rock characteristics, and brine chemistry. According to the reviewed studies (Table 3), H_2, CH_4, CO_2, and N_2 exhibit markedly different mobilities, displacement patterns, and trapping efficiencies in both sandstone and carbonate formations. H_2 consistently demonstrates very high mobility, primarily due to its low viscosity and density (Juez-Larré et al., 2023; Zhong et al., 2024; Song et al., 2025). These properties enable rapid migration but also increase the risk of viscous fingering and gravity override when displacing denser fluids. In Berea sandstone, Lysyy et al. (2022) found that H_2 had lower relative permeability than N_2 and a residual gas saturation of 0.36, indicating moderate capillary trapping compared to N_2. Pore-scale imaging by Zhang et al. (2023) in Bentheimer sandstone confirmed that H_2's low viscosity promotes channeling, leading to low initial saturation after drainage (0.44) and very low residual saturation after imbibition (0.11), with dissolution further contributing to storage losses.

Table 3—Summary of experimental and modeling studies on gas transport and trapping behavior for H_2, CO_2, CH_4, and N_2 in various rock types.

Study	Gas	Rock	Temperature (K)	Pressure (MPa)	Brine total salinity (wt%)	Gas transport and trapping
Al-Yaseri et al. (2022)	H_2	Fontainebleau sandstone	298	≈0.4	3 (NaCl)	Displaced only a small amount of brine; uniform displacement during imbibition. ~1.7% residual saturation at 2 mL/min; very low trapping even when capillary number is matched via N_2 test.
	N_2					Much higher displacement of brine than H_2 under same conditions. ~16.9% residual saturation at 2 mL/min; ~7.8% when capillary number matched to H_2.
Lysyy et al. (2022)	H_2	Berea sandstone	303	3	5 (50% NaCl + 50% CsCl)	lower relative permeability than N_2. Residual gas saturation = 0.36.
	N_2					Higher relative permeability than H_2.
Alanazi et al. (2023)	H_2	organic-rich carbonate	333	0.1 - 10		Low adsorption capacity (0.18–0.5 mol/Kg); higher adsorption with higher TOC.
	CO_2					High adsorption capacity (0.01–12 mol/kg).
	CH_4					Moderate adsorption capacity (0.2–1 mol/kg).
	N_2					Moderate adsorption capacity (0.73 mol/kg).
Al-Shafi et al. (2023)	CH_4					High mobility.
	H_2					Very high mobility.
	CO_2					Moderate mobility.
Juez-Larré et al. (2023)	H_2	Carbonate and sandstone	318.15-390.15	18-39.3		Higher mobility due to lower viscosity and density.
	CH_4					Lower mobility due to higher viscosity and density.
Saeed et al. (2023)	H_2	Sandstone	353-393	10.133-50.663	2.922-29.22 (NaCl)	High mobility; very low viscosity and density; losses <2% over 30 yrs
	CO_2					Moderate mobility; substantial dissolution & reaction; up to ~72% loss over 30 yrs
Zhang et al. (2023)	H_2	Bentheimer sandstone	298	1	3.4 (KI)	Higher mobility due to lower viscosity (0.88×10^{-5} Pa·s), more channeling, lower initial saturation (0.44) after drainage. Lower residual saturation (0.11) after imbibition; dissolution contributes to loss.
	N_2					Lower mobility due to higher viscosity (1.76×10^{-5} Pa·s), higher initial saturation (0.83) after drainage. Higher residual saturation (0.36) after imbibition; limited dissolution.
Du et al. (2024)	CO_2					Low mobility due to high viscosity.
	CH_4					High mobility.
Zhong et al. (2024)	CO_2					Moderate viscosity vs H_2; less prone to fingering than H_2.
	H_2					Very high mobility (low μ) → viscous fingering risk due to fluid mobility contrast, and gravity override due to fluid densities differences.
Song et al. (2025)	N_2		298.15	4.5-10		Moderate adsorption capacity (0.73 mol/kg).
	N_2					Low mobility.
	CH_4					Moderate mobility.

In reactive systems, H_2 exhibits minimal geochemical loss compared to CO_2. Saeed et al. (2023) reported <2% H_2 loss over 30 years in sandstone, whereas CO_2 experienced up to ~72% loss from dissolution and mineral reactions under similar conditions. In organic-rich carbonates, H_2 adsorption is low (0.18–0.5 mol/kg) but increases with total organic content (Alanazi et al., 2023), suggesting surface sorption plays a minor role in containment except in formations with a high TOC.

CO_2 generally shows moderate mobility, due to its higher viscosity compared to H_2 but lower than CH_4 under some conditions (Al-Shafi et al., 2023; Du et al., 2024). Its reduced mobility relative to H_2 makes it less prone to fingering and gravity override (Zhong et al., 2024), improving sweep efficiency in homogeneous formations. However, CO_2's strong solubility in brines and reactivity with carbonate minerals lead to substantial mass losses over time (Saeed et al., 2023). In adsorption studies, CO_2 shows a high adsorption capacity up to 12 mol/kg (Alanazi et al., 2023), which may enhance storage security in organic-rich formations.

CH_4 typically has moderate to high mobility, with higher viscosity and density than H_2 but lower than CO_2 in some cases (Juez-Larré et al., 2023; Du et al., 2024; Song et al., 2025). CH_4's adsorption capacity in organic-rich carbonates is moderate (0.2–1 mol/kg) (Alanazi et al., 2023), potentially enhancing storage security in unconventional reservoirs. Unlike CO_2, CH_4 has lower solubility in brines, reducing dissolution losses, but retains some fingering risk in highly heterogeneous media.

N_2 displays a wide range of mobilities depending on formation conditions, generally being less mobile than H_2 due to its higher viscosity (Zhang et al., 2023; Song et al., 2025). In Bentheimer sandstone, N_2 showed higher initial saturation after drainage (0.83) and higher residual saturation after imbibition (0.36) than H_2 (Zhang et al., 2023), indicating stronger capillary trapping potential. In Fontainebleau sandstone, Al-Yaseri et al. (2022) found that N_2 displaced substantially more brine than H_2 under identical conditions, with much higher trapped residual saturations (~16.9% vs 1.7%, reinforcing its suitability for long-term gas storage where immobile trapping is desired.

The overall trends suggest that H_2 offers the highest injectivity and fastest displacement while CO_2 benefits from moderate mobility and better sweep efficiency. CH_4 provides balanced mobility and storage stability (if used as a cushion gas), while N_2 offers lower mobility, potentially limiting injectivity.

Studies investigating the relationship between initial and residual gas saturation for different rock types, brine compositions, and pressure–temperature conditions are compared in Figure 4. For H_2, Al-Yaseri et al. (2022) observed low residual gas saturation values in Fontainebleau sandstone (0.017 at an initial gas saturation of 0.041), whereas in Gosford sandstone higher values were reported, with residual saturations exceeding 0.44 (Al-Yaseri et al., 2024). Similarly, Bentheimer sandstone exhibited low residual H_2 saturations of 0.11 at initial saturations of 0.44 (Zhang et al., 2023), while Berea sandstone showed a wide range from 0.0204 to 0.385 depending on sample heterogeneity (Boon & Hajibeygi, 2022). For N_2, residual saturations varied from as low as 0.078 in Fontainebleau sandstone (Al-Yaseri et al., 2022) to as high as 0.36 in Bentheimer sandstone (Zhang et al., 2023). For CO_2 a wide range of values have been reported with values ranging from 0.075–0.33 across water-wet sandstones and carbonates (El-Maghraby & Blunt, 2013; Andrew et al., 2014; Rahman et al., 2016). The influence of wettability was evident in mixed-wet carbonate systems, where Al-Menhali and Krevor (2016) reported reduced CO_2 residual saturation (0.15) compared to water-wet conditions (0.24). Additionally, variations in permeability impacted trapping efficiency, with Mouallem et al. (2025) demonstrating higher residual CO_2 saturation in Indiana limestone with intermediate permeability (27.3 mD) compared to very low (5.26 mD) and very high permeability (680 mD) samples. Collectively, these findings indicate that gas type, mineralogy, brine chemistry, wettability, and pore structure jointly govern gas trapping behavior in geological formations, with H_2 generally showing lower trapping capacity compared to CO_2 but only under strongly water-wet conditions, with these values generally lower than for N_2. However, as evident in Figure 4, there is a wide scatter of data and further studies are required to reach definitive conclusions.

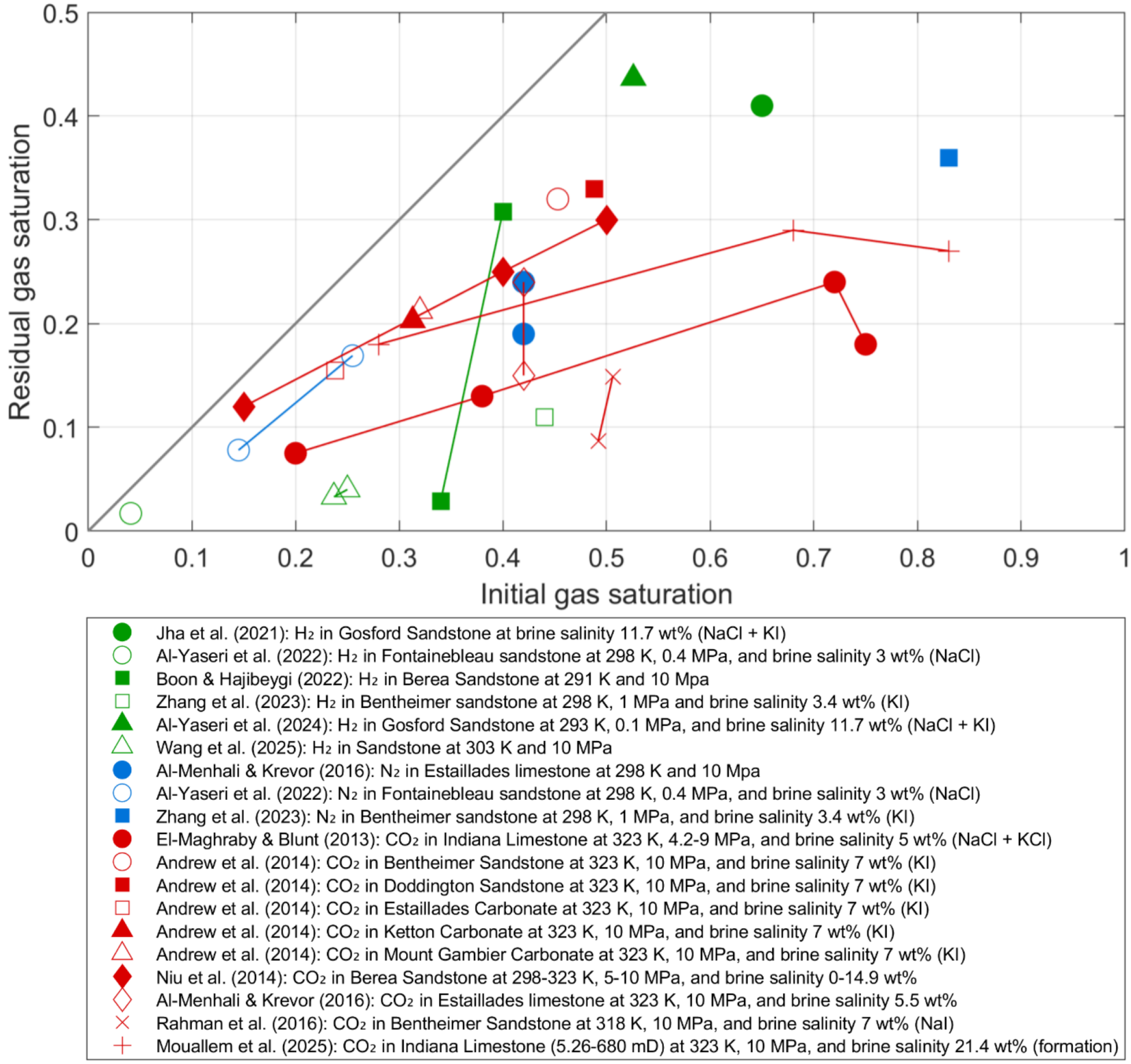

Figure 4—Initial vs residual gas saturation of different studies where green is H_2, blue is N_2 and red is CO_2. The diagonal line represents 100% trapping.

Conclusions

This research underscores the distinct subsurface multiphase flow and transport behaviors of H_2, CO_2, CH_4, and N_2, revealing that differences in diffusion, wettability, and pore-scale dynamics govern their transport and trapping mechanisms. H_2's high diffusion rate and rapid Ostwald ripening effect leads to a reduction in hysteresis and less H_2 trapping. CO_2 exhibits a dual influence—modifying wettability and undergoing Ostwald ripening—while offering moderate mobility. CH_4 is expected to be affected by Ostwald ripening due to its higher diffusion coefficient. N_2 has lower mobility which may constrain injectivity. Overall, the extent and causes of hysteresis, wettability change, mobility, and trapping differ sharply among these gases, driven by variations in interfacial properties and physicochemical processes. For UHS, these findings highlight the need for an optimized cycling schedule—to mitigate migration risks and enhance recovery. In contrast, the more stable behavior and lower mobility of CO_2 and N_2 behave better as cushion gases than CH_4 for long-term gas storage. These insights provide a framework for tailoring gas-specific storage strategies to maximize security, efficiency, and long-term performance in diverse geological settings. However, the experimental evidence collected to date shows wide variability with few good-quality comparative studies. Further work is required to provide more robust and accurate assessments of capillary trapping and multiphase flow properties.

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